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Machine learning–based prediction of high hospital charges after lung transplantation in chronic obstructive pulmonary disease patients

Qunzhe Ding^{1†}, Tianyu Sun^{2†}, Zihao Zhang³, Yuhong Liu², Mengli Zheng², Pengtao Bao² and Xinxin Zhang^{2*}

Abstract

Background Lung transplantation is an important therapeutic option for patients with end-stage chronic obstructive pulmonary disease. However, post-transplantation hospitalizations are associated with substantial cost variation. Accurately predicting high in-hospital expenditures is essential for optimizing clinical management and healthcare resource allocation.

Methods This study utilized the Nationwide Inpatient Sample database (2005–2014) to identify adult patients who underwent lung transplantation. After rigorous data preprocessing, seven post-discharge or manually coded variables were excluded to prevent potential information leakage. High-cost hospitalizations were defined as those with total hospital charges falling within the top 20% (≥ 80 th percentile) of the overall cost distribution. Feature selection was performed using a two-stage process combining correlation analysis and Recursive Feature Elimination. Six machine learning algorithms were trained through ten-fold cross-validation with optimized hyperparameters, and model performance was compared across seven evaluation methods. The optimal model was further interpreted to identify the key features associated with high-cost hospitalizations.

Results The Gradient Boosting Machine model achieved the most robust predictive performance across all evaluation metrics, with consistently high accuracy (0.816), F1-score (0.6773), and area under the curve (AUC = 0.8078). To further interpret the model's decision mechanism, Shapley Additive Explanations analysis was applied to quantify each feature's contribution to high-cost prediction. The results revealed that hospital region, mechanical ventilation, acute kidney injury, lung transplant type, and complications of the transplanted lung were the most influential predictors of high-cost hospitalizations. Notably, hospitalizations in the northeast region were associated with the highest predicted costs, indicating substantial regional variation in post-transplant expenditure patterns.

Conclusions This study developed a transparent machine learning method for predicting high-cost hospitalizations after lung transplantation. The Gradient Boosting Machine model demonstrated optimal predictive performance, and

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Shapley analysis identified key clinical and regional factors driving hospital expenditures, providing evidence to guide cost management and clinical decision-making.

Keywords Lung transplantation, Chronic obstructive pulmonary disease, Healthcare expenditure, Machine learning, Cost prediction

Introduction

Chronic obstructive pulmonary disease (COPD) represents a significant global health burden, characterized by progressive airflow limitation and respiratory symptoms [1, 2]. For patients with end-stage COPD, lung transplantation (LT) emerges as a potentially life-extending therapeutic option, offering improved quality of life and functional status [3, 4]. However, LT entails not only clinical considerations but also substantial financial implications, with total hospital charges encompassing a diverse array of expenses spanning preoperative assessments, surgical interventions, postoperative care, and management of complications.

The ability to accurately predict total hospital charges following LT in COPD patients holds profound implications for healthcare providers, payers, and patients. Such predictions facilitate informed decision-making, resource allocation, and financial planning, thereby enhancing the efficiency and sustainability of LT programs. Traditional approaches to estimating hospital charges often rely on simplistic models that overlook the intricate interplay of patient-specific factors, disease characteristics, and procedural variables. In recent years, the advent of machine learning techniques has revolutionized predictive analytics in healthcare, offering unparalleled capabilities for discerning complex patterns and relationships. Machine learning has been used to predict satisfaction [5], complications [6], and readmission [7], which help to improve patient outcomes and cost effectiveness.

The current study aims to develop a predictive model using machine learning for the prediction of high total hospital charges. In this study, patients with high in-hospital costs were defined as those whose total hospital charges fell within the top 20% of the overall cost distribution, that is, whose total charges exceeded the 80th percentile of all patients in the dataset, following the percentile-based definitions commonly adopted in prior high-cost prediction studies [8]. Our primary objective is to provide clinicians, healthcare administrators, and policymakers with a model for anticipating total hospital charges in COPD patients undergoing LT. Also, with the explanation of the best-performed model, the results reveal the most important factors and demonstrate how these factors will influence healthcare costs. Our predictive model is designed to support clinical decision-making, improve resource distribution, and strengthen the economic viability of lung transplantation programs focused on managing COPD.

Materials and methods

Data source

The Nationwide Inpatient Sample (NIS) database was used to identify patients undergoing LT from January 2005 to December 2014. NIS is a part of the Healthcare Cost and Utilization Project (HCUP) by the Agency for Healthcare Research and Quality [8, 9]. Information on database content, access procedures, and documentation is publicly available through the HCUP Central Distributor (<https://cdors.ahrq.gov/databases>). Variables in the database mainly included demographics, hospital-associated, clinical, and resource utilization-related characteristics.

Ethical approval

The NIS contained deidentified patient information, thus the study was exempt from approval of the Institutional Review Board at the Chinese PLA General Hospital. Informed consent to participate from each participant was waived as assessed by the Chinese PLA General Hospital Institutional Review Board.

Data collection and preprocessing

A total of 1,177 adult (age > 18 years) hospital admissions with procedural codes indicating lung transplantation (ICD-9-CM 33.51 for unilateral and 33.52 for bilateral procedures) were initially retrieved from the database. The original dataset comprised 36 variables capturing demographic, socioeconomic, hospital, and clinical information. To address the potential issue of information leakage raised during the review process, seven variables that were determined only after discharge or manually coded post-hospitalization were excluded, including APR-DRG Risk of Mortality, APR-DRG Severity of Illness, Length of Stay, In-hospital Mortality, Discharge Disposition, Discharge Year, and Bilateral Lung Transplant.

The final analytical dataset contained 31 variables, of which 29 were retained as predictive features. These variables were organized into three broad categories. The first category, demographic and socioeconomic characteristics, included Age at Admission, Sex, Race, Insurance Type, Elective Admission, and Zip-Code Income Quartile. These variables captured basic patient profiles and social determinants potentially associated with healthcare utilization and cost. The second category, hospital attributes, reflected institutional characteristics that may influence treatment decisions and resource

use, including Hospital Region and Hospital Bed Size. The third category, clinical and comorbidity features, encompassed a comprehensive set of pre-admission or contemporaneous medical conditions relevant to lung transplantation outcomes and hospitalization costs. These features comprised Alcohol-Related Disorders, Congestive Heart Failure, Chronic Pulmonary Disease, Coagulopathy, Hypertension (Complicated), Fluid and Electrolyte Disorders, Obesity, Pulmonary Circulation Disorders, Weight Loss, Current Smoking, Unilateral Lung Transplant, Liver Disease, Acute Kidney Injury, Sepsis, Mechanical Ventilation, ECMO, Complications of Transplanted Lung, Respiratory Failure, and *Clostridioides difficile* Infection. Each record additionally contained a Patient Identifier serving as a unique patient-level key, and Total Hospital Charges designated as the outcome variable.

To facilitate predictive modeling, Total Hospital Charges were further processed to construct a binary outcome variable representing high-cost hospitalizations. Prior to analysis, 12 admissions with missing charge values were excluded to ensure accuracy in percentile estimation. The 80th percentile of the remaining total charge distribution (\$ 600,223.40) was applied as the

cutoff threshold. Admissions with total hospital charges greater than or equal to this value were classified as the high-cost group, whereas those below the threshold were assigned to the low-cost group. According to this criterion, 232 admissions were categorized as high-cost cases and 927 admissions as low-cost cases. This percentile-based approach enabled a consistent and interpretable separation of the top 20% of admissions in terms of total hospitalization cost. As shown in Fig. 1, the boxplot demonstrates the distributional differences in total hospital charges between the high- and low-cost groups.

Two categorical variables, Race and Zip-Code Income Quartile, contained missing values in 198 and 28 records, respectively. To maintain data integrity and preserve the multivariate relationships among predictors, missing values were imputed using the Multiple Imputation by Chained Equations (MICE) framework. Within this framework, a Gradient Boosting Classifier was implemented as the predictive model for imputing multi-class categorical variables. This tailored imputation strategy allowed flexible modeling of non-linear relationships and interactions among predictors, resulting in robust and accurate estimation of missing categories across the dataset.

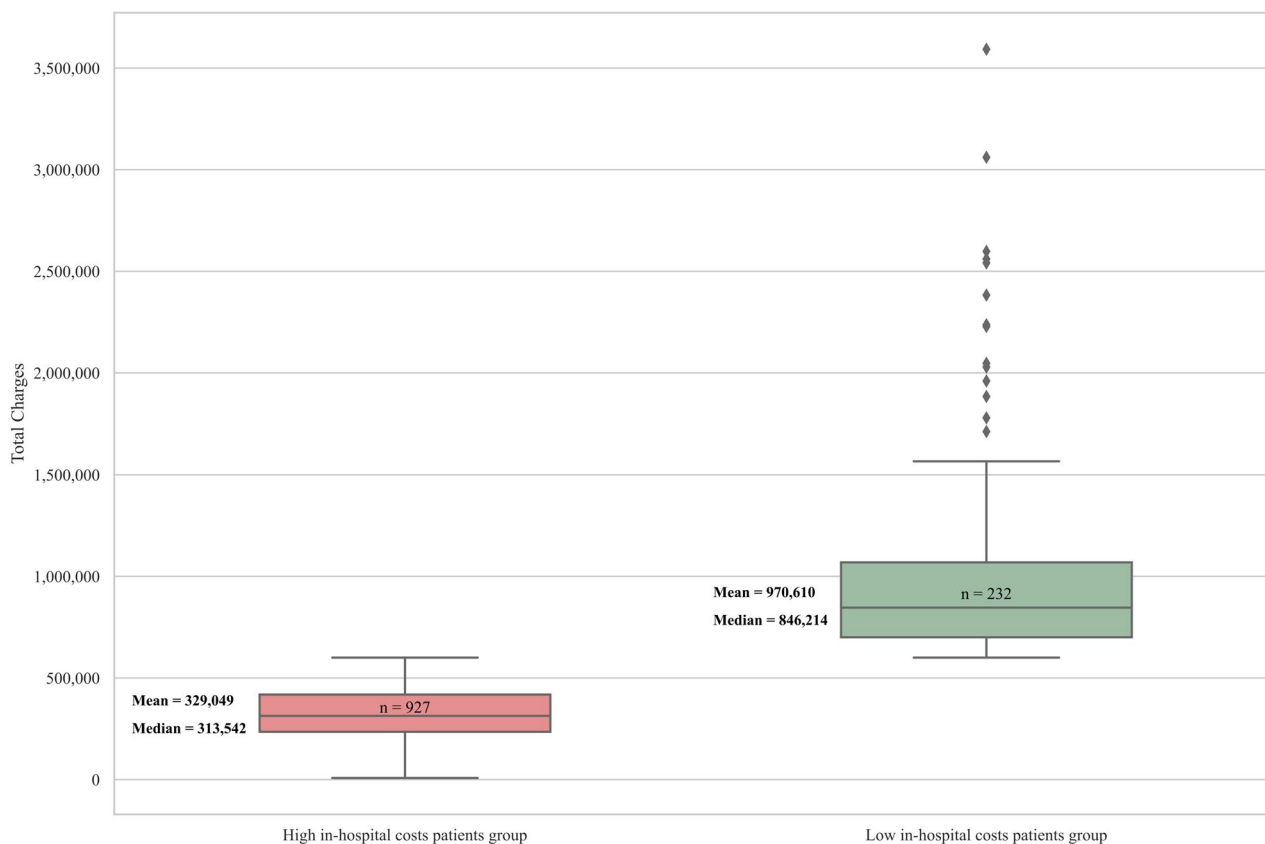


Fig. 1 Distribution of total hospital charges between high- and low-cost hospitalization groups. The boxplot compares total hospital charges between the two groups. The high-cost group ($n=232$) had higher median and mean charges (\$846,214 and \$970,610) than the low-cost group (\$313,542 and \$329,049). Several outliers were noted among high-cost admissions, indicating large variability in expenditures

Statistical analysis

The experimental analyses were conducted using Python version 3.10.9. Data preprocessing and manipulations were performed utilizing libraries such as pandas, NumPy, and Scikit-Learn, among others. Patients were stratified into training and test cohorts in a 7:3 ratio, with approximately 70% of the dataset dedicated to model training and the remaining 30% for validation. To find optimal fine-tune hyperparameters, ten-fold cross-validation was employed during the training process. The predictive accuracy of the GBM algorithm was compared against a suite of established machine learning algorithms including Random Forest (RF), XGBoost, Catboost, and LightGBM. Model performance was assessed through receiver operating characteristic (ROC) curve analysis, utilizing the area under the ROC curve (AUC) and confusion matrices as the primary evaluative metrics.

The Gradient Boosting Machine (GBM) algorithm, developed by Friedman [10], enhances the decision tree modeling process through an iterative method to reduce prediction errors in models. It minimizes the loss function by iteratively constructing decision trees, where each iteration adds a new tree that fits the residuals of all predictions made by the trees in the previous iterations. This method allows for gradual improvement of the model by accurately adjusting for prediction errors, thus enhancing the overall predictive accuracy. The essence of the GBM algorithm lies in employing gradient descent to guide the construction of decision trees in each round, aiming to reduce the model's loss as efficiently as possible. Additionally, GBM supports the subsampling of input features, which aids in enhancing the model's generalization capability, increasing its diversity, and preventing overfitting. The core principle of GBM is to optimize the prediction model iteratively based on a given set of feature vectors and their corresponding output categories, thereby reducing prediction errors. During the training process, the model is fine-tuned by adjusting parameters such as the number of trees, their depth, and the learning rate. This adjustment balances the model's complexity and training effectiveness, ensuring the model possesses sufficient fitting capacity without succumbing to overfitting. The formula of GBM can be represented as:

$$y_i^{(t)} = \sum_{k=1}^t f_k(x_i), f_k \in F,$$

Where each function f_k represents an individual decision tree added to the model at iteration k , and F is the space of all possible trees. This summation shows how the final prediction y_i at iteration t is composed of the sum of the predictions from all trees up to that point.

Following model optimization, predictive performance was comprehensively assessed using a suite of evaluation metrics: Accuracy, Precision, Recall, F1 Score, Area Under the ROC Curve (AUC), Precision–Recall (P–R) curves, and calibration performance. This multi-metric evaluation ensured a robust and reliable comparison of model performance.

SHAP (Shapley Additive Explanations) [11], as a powerful framework for interpreting machine learning models by attributing prediction outcomes to individual features at both global and local levels. Grounded in cooperative game theory, SHAP employs Shapley values to equitably allocate contribution scores among features, thereby providing an interpretable and consistent measure of each feature's influence on model predictions. At the global level, SHAP facilitates the ranking of feature importance and reveals complex feature interactions, offering a comprehensive understanding of how the model makes decisions across the entire dataset. At the local level, SHAP dependence plots visualize how variations in a single feature influence individual prediction, while SHAP interaction plots illustrate the joint impact of two features on the model's output. In this study, SHAP was employed to interpret the best-performing model, with the goal of identifying the most influential features and exploring their marginal effects and on prediction outcomes. This interpretability analysis offered deeper insights into the model's internal reasoning and enhanced the transparency of the predictive framework.

Results

Feature selection

In this study, feature selection was performed through a two-stage process to remove redundant predictors and retain those with the highest explanatory value for high-cost hospitalizations. In the first stage, a Pearson correlation heatmap was generated to evaluate pairwise linear relationships among all input variables. As illustrated in Fig. 2A, this analysis revealed generally low to moderate inter-feature correlations, with most absolute correlation coefficients below 0.4. The absence of strong collinearity indicated that all variables could be retained for further modeling without risk of redundancy or inflated variance.

Following this initial filtering, Recursive Feature Elimination (RFE) was applied as the second-stage selection method to further refine the feature set. RFE iteratively fits a predictive model, in this case a Random Forest classifier that was adopted as a robust and widely used baseline for tabular feature ranking, and ranks variables according to their relative importance, successively removing the least informative predictors until an optimal subset is obtained. Based on this procedure, Liver disease and *Clostridioides difficile* Infection were identified as the least contributory features and were therefore

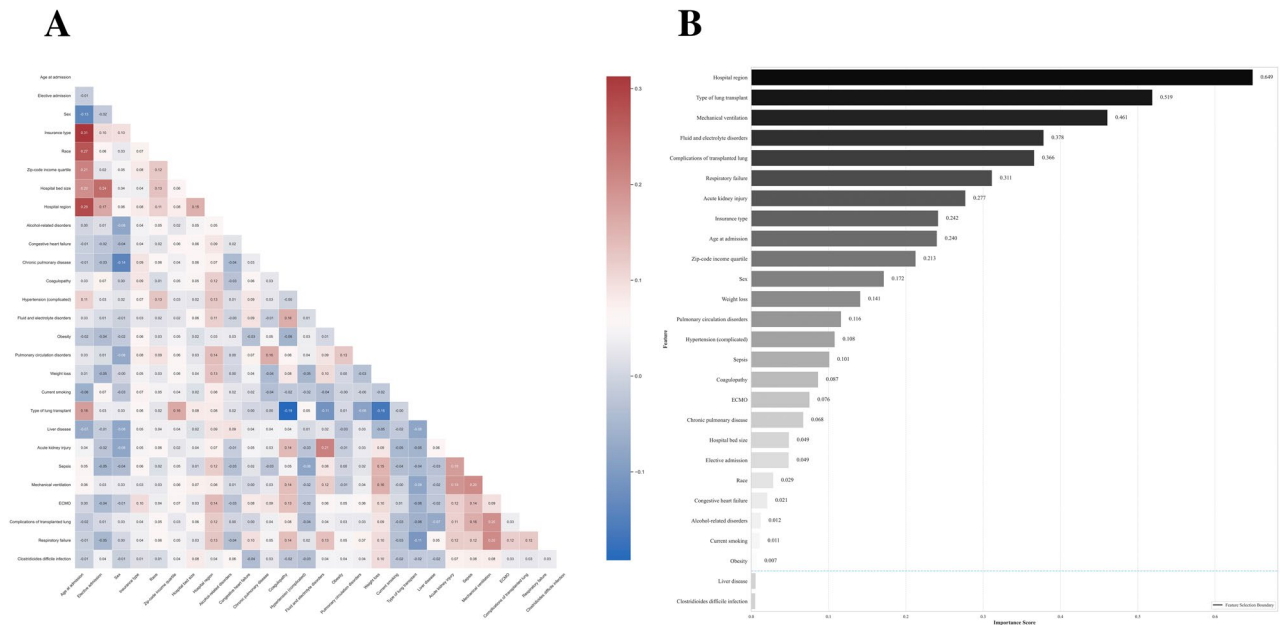


Fig. 2 Correlation assessment and feature selection for high-cost hospitalization modeling. **(A)** Pearson correlation heatmap showing generally low inter-feature correlations ($|r| < 0.4$), indicating no significant multicollinearity among predictors. **(B)** Recursive Feature Elimination (RFE) using a Random Forest classifier to rank variable importance

excluded to improve model parsimony and reduce feature redundancy, as illustrated in Fig. 2B. After completing both selection stages, a total of 27 predictive variables were retained for subsequent model development and evaluation.

Baseline characteristics

A total of 1,159 patients were included in the analysis, and their baseline characteristics are presented in Table 1. Group comparisons were performed using the chi-square test for categorical variables and the Mann–Whitney U test for continuous variables. Compared with patients in the low-cost group, those with high hospital charges were significantly more likely to have undergone bilateral lung transplantation (77.59% vs. 57.07%, $P < 0.001$) and exhibited higher rates of acute kidney injury (41.81% vs. 16.40%, $P < 0.001$), sepsis (16.81% vs. 2.91%, $P < 0.001$), mechanical ventilation (57.33% vs. 27.72%, $P < 0.001$), ECMO use (11.21% vs. 1.29%, $P < 0.001$), and complications of the transplanted lung (53.88% vs. 31.18%, $P < 0.001$). In addition, chronic pulmonary disease, coagulopathy, fluid and electrolyte disorders, pulmonary circulation disorders, weight loss, and *Clostridioides difficile* infection were more prevalent among high-cost patients (all $P < 0.05$). A marked geographic disparity was observed, with a substantially higher proportion of high-cost admissions in hospitals located in the North-east region (38.36% vs. 10.36%, $P < 0.001$), whereas demographic characteristics such as sex, race, and obesity

showed no significant differences between the two cost groups ($P > 0.05$).

Model comparison

Consistent with the procedures described in the Methods section, all models were trained using ten-fold cross-validation with optimized hyperparameters. A comprehensive summary of the hyperparameter search ranges and corresponding optimal parameter settings is provided in Supplementary Table 1. Upon completing cross-validation and parameter tuning, we evaluated and compared the predictive performance of all six models, as summarized in Table 2. Overall, the performance of the six models was broadly comparable. Among the gradient-boosting algorithms, GBM achieved the highest accuracy (0.8164) and recall (0.6657), resulting in the best overall F1-score (0.6773). CatBoost produced the same AUC as GBM (0.8078) but yielded slightly lower recall and F1, while maintaining the highest precision (0.6971). XGBoost showed very similar performance (AUC=0.8043) with balanced accuracy and recall, and LightGBM achieved stable but marginally lower results (AUC=0.7967). In contrast, Logistic Regression and Random Forest underperformed across all indicators (AUC=0.7771 and 0.7636, respectively), reflecting limited capacity to capture accurate performance.

To further validate model performance, this study generated AUC-ROC, precision–recall (PR), and calibration curves (Fig. 3A–C). Overall, all models showed comparable discrimination, with GBM maintaining consistently

Table 1 Baseline characteristics of patients

Variable	Category	Patients with low hospital charges (927, 79.98%)	Patients with high hospital charges (232, 20.02%)	P-Value
Sex	Male	495 (53.40%)	130 (56.03%)	0.518
	Female	432 (46.60%)	102 (43.97%)	
Insurance type	Medicare	449 (48.44%)	141 (60.78%)	0.006
	Private insurance	396 (42.72%)	72 (31.03%)	
	Medicaid	44 (4.75%)	12 (5.17%)	
	Self-pay/Other	38 (4.10%)	7 (3.02%)	
Race	White	835 (90.08%)	203 (87.50%)	0.341
	Black	52 (5.61%)	19 (8.19%)	
	Hispanic/Native American/ Others	40 (4.31%)	10 (4.31%)	
Zip-code income quartile	0-25th	197 (21.25%)	49 (21.12%)	0.362
	26th-50th	236 (25.46%)	66 (28.45%)	
	51th-75th	261 (28.16%)	71 (30.60%)	
	76th-100th	233 (25.13%)	46 (19.83%)	
Hospital bed size	Small	11 (1.19%)	3 (1.29%)	0.878
	Medium	41 (4.42%)	12 (5.17%)	
	Large	875 (94.39%)	217 (93.53%)	
Hospital region	South	319 (34.41%)	59 (25.43%)	< 0.001
	West	252 (27.18%)	54 (23.28%)	
	Midwest	260 (28.05%)	30 (12.93%)	
	Northeast	96 (10.36%)	89 (38.36%)	
Alcohol-related disorders	No	911 (98.27%)	226 (97.41%)	0.555
	Yes	16 (1.73%)	6 (2.59%)	
Congestive heart failure	Absent	880 (94.93%)	215 (92.67%)	0.236
	Present	47 (5.07%)	17 (7.33%)	
Chronic pulmonary disease	Absent	638 (68.82%)	143 (61.64%)	0.044
	Present	289 (31.18%)	89 (38.36%)	
Coagulopathy	Absent	735 (79.29%)	166 (71.55%)	0.015
	Present	192 (20.71%)	66 (28.45%)	
Hypertension (complicated)	No	547 (59.01%)	146 (62.93%)	0.310
	Yes	380 (40.99%)	86 (37.07%)	
Fluid and electrolyte disorders	No	531 (57.28%)	88 (37.93%)	< 0.001
	Yes	396 (42.72%)	144 (62.07%)	
Obesity	No	886 (95.58%)	221 (95.26%)	0.974
	Yes	41 (4.42%)	11 (4.74%)	
Pulmonary circulation disorders	Absent	701 (75.62%)	154 (66.38%)	0.005
	Present	226 (24.38%)	78 (33.62%)	
Weight loss	No	841 (90.72%)	168 (72.41%)	< 0.001
	Yes	86 (9.28%)	64 (27.59%)	
Current smoking	Non-smoker	904 (97.52%)	231 (99.57%)	0.088
	Current smoker	23 (2.48%)	1 (0.43%)	
Type of lung transplant	Bilateral transplant	529 (57.07%)	180 (77.59%)	< 0.001
	Unilateral transplant	398 (42.93%)	52 (22.41%)	
Liver disease	Absent	913 (98.49%)	230 (99.14%)	0.658
	Present	14 (1.51%)	2 (0.86%)	
Acute kidney injury	No	775 (83.60%)	135 (58.19%)	< 0.001
	Yes	152 (16.40%)	97 (41.81%)	
Sepsis	No	900 (97.09%)	193 (83.19%)	< 0.001
	Yes	27 (2.91%)	39 (16.81%)	
Mechanical ventilation	No	670 (72.28%)	99 (42.67%)	< 0.001
	Yes	257 (27.72%)	133 (57.33%)	
ECMO	No	915 (98.71%)	206 (88.79%)	< 0.001
	Yes	12 (1.29%)	26 (11.21%)	

Table 1 (continued)

Variable	Category	Patients with low hospital charges (927, 79.98%)	Patients with high hospital charges (232, 20.02%)	P-Value
Complications of transplanted lung	No	638 (68.82%)	107 (46.12%)	< 0.001
	Yes	289 (31.18%)	125 (53.88%)	
Respiratory failure	No	597 (64.40%)	100 (43.10%)	< 0.001
	Yes	330 (35.60%)	132 (56.90%)	
Clostridioides difficile infection	No	909 (98.06%)	220 (94.83%)	0.011
	Yes	18 (1.94%)	12 (5.17%)	
Elective admission	Non-elective	708 (76.38%)	186 (80.17%)	0.253
	Elective	219 (23.62%)	46 (19.83%)	
Age at admission		59.51 ± 7.54	60.56 ± 8.57	0.002

Table 2 The performance comparison of machine learning models

	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression	0.8017	0.6780	0.6247	0.6406	0.7771
Random Forest	0.7931	0.6580	0.6086	0.6224	0.7636
GBM	0.8164	0.6944	0.6657	0.6773	0.8078
XGBoost	0.8075	0.6923	0.6497	0.6649	0.8043
CatBoost	0.8103	0.6971	0.6248	0.6437	0.8078
LightGBM	0.7989	0.6757	0.6390	0.6521	0.7967

strong ROC performance among the gradient-boosting algorithms (Fig. 3A). In the PR analysis, GBM achieved the largest area under the curve (AP=0.5261), indicating a stronger ability to identify true high-cost cases (Fig. 3B). Calibration plots showed that CatBoost and LightGBM were best aligned with the ideal reference line, while GBM also demonstrated good calibration with a low Brier score (0.1360) (Fig. 3C). The confusion matrix of the GBM model (Fig. 3D) confirmed balanced classification performance, supporting its stable trade-off between sensitivity and specificity.

Taken together, GBM exhibited the most favorable overall performance across discrimination, calibration, and classification metrics, and was therefore identified as the optimal predictive model for this study.

Model interpretability analysis

To interpret the prediction logic of the model, SHAP analysis was employed to quantify and visualize the contribution of each clinical feature using two complementary plots: the SHAP feature importance plot and the SHAP summary plot (Fig. 4). The feature importance plot ranks predictors by their mean absolute SHAP values, representing their overall contribution to the model's predictions. As shown in Fig. 4A, the five most influential factors were hospital region, mechanical ventilation, acute kidney injury, type of lung transplant, and complications of transplanted lung. These variables together highlight the combined effects of institutional context, perioperative complications, and transplant characteristics on hospital charges. The SHAP summary plot

(Fig. 4B) further illustrates how variations in feature values influence the model output. Higher severity-related conditions such as mechanical ventilation and acute kidney injury (shown in red) were associated with increased SHAP values, indicating a greater likelihood of high-cost hospitalization. Conversely, lower feature values (blue) for features such as Hypertension contributed to reduced cost predictions. Overall, the SHAP analysis demonstrates that both clinical acuity and hospital-level heterogeneity are key influencers of elevated costs in lung-transplant patients.

However, some features exhibited heterogeneous SHAP value distributions, suggesting that their effects may depend on complex contextual interactions rather than a single directional trend. To further elucidate how these key predictors contribute to individual-level predictions, the study subsequently employed SHAP local interpretation plot for a more fine-grained interpretability analysis.

To further elucidate how individual clinical and contextual features influenced the model's predictions, SHAP dependence plots were generated for the six most influential variables (Fig. 5A–F). These local interpretations illustrate how changes in specific feature values affected the predicted probability of high hospital charges.

As shown in Fig. 5A, hospital region exhibited a pronounced spatial gradient, with admissions in the Northeast associated with substantially higher SHAP values, suggesting that regional cost heterogeneity plays one of the most decisive roles in shaping financial outcomes. Figure 5B demonstrates that mechanical ventilation markedly increased SHAP values, indicating a strong positive contribution to cost prediction, consistent with the intensive care requirements of ventilated patients. Similarly, Fig. 5C shows that the occurrence of acute kidney injury significantly elevated SHAP values, reflecting its substantial economic burden due to extended hospitalization and treatment complexity. In Fig. 5D, the type of lung transplant displayed a clear dichotomous trend, with unilateral transplantation corresponding to higher SHAP values compared with bilateral transplantation,

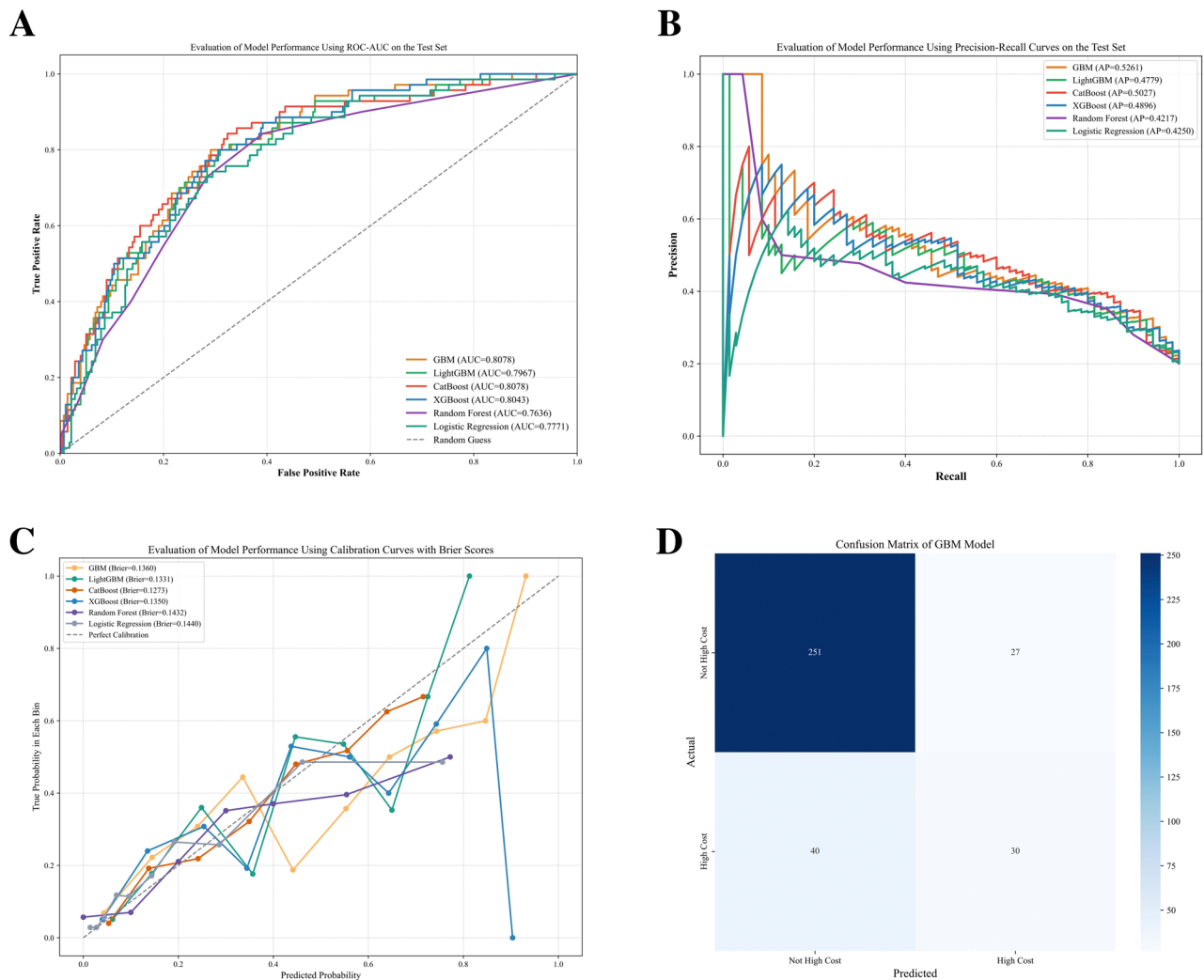


Fig. 3 Comparison of model discrimination, precision-recall performance, calibration, and classification performance in the test cohort. **(A)** Receiver operating characteristic (ROC) curves of the six prediction models. **(B)** Precision–recall (PR) curves of the six prediction models. **(C)** Calibration plots comparing predicted and observed probabilities. **(D)** Confusion matrix of the GBM model on the test set

implying greater predicted costs potentially related to perioperative or postoperative management differences. Figure 5E indicates that patients with complications of the transplanted lung were associated with increased SHAP values, highlighting the financial impact of postoperative complications. Finally, Fig. 5F reveals that the presence of fluid and electrolyte disorders contributed positively to cost predictions, underscoring the importance of metabolic imbalances as cost-amplifying factors.

Taken together, these results indicate that regional disparities exerted the strongest overall influence on hospitalization costs, exceeding the effects of individual clinical severity and highlighting the structural nature of cost heterogeneity within the U.S. lung transplant care system during the study period.

Discussion

In the current study, we included 1159 cases and identified eleven significant features impacting hospital charges, including severity of illness, risk of mortality, hospital region, lung transplant type, weight loss, electrolyte disorders, age, coagulopathy, primary payer, pulmonary circulation disorders, and chronic lung disease. The GBM model emerged as the most effective predictive model, exhibiting the highest area under the curve (AUC) of 80.78%. The GBM model demonstrated robust performance in the test set, with an accuracy of 0.8164, precision of 0.6944, recall of 0.6657, and F1 score of 0.6773. These metrics underscore the model's efficacy in distinguishing between patients with high and low hospital charges, reflecting its potential utility in a clinical setting.

Utilizing the SHAP method, we identified the top five features contributing to high hospital charges: hospital

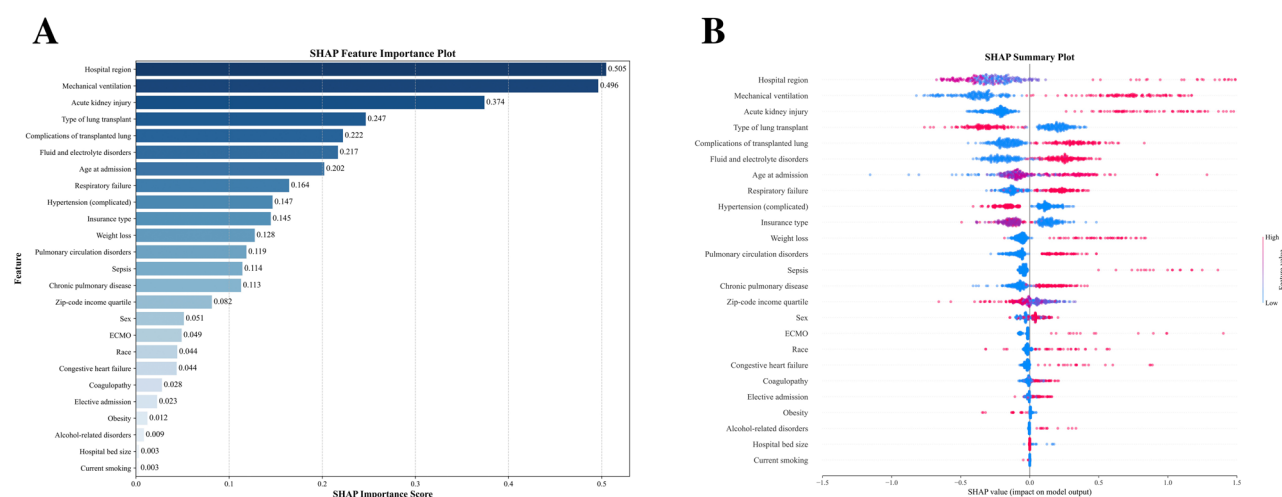


Fig. 4 SHAP-based global interpretation of the GBM model. **(A)** SHAP summary plot illustrating how individual feature values influence the model output—red indicates higher values, and blue indicates lower values. **(B)** Feature importance plot for comorbidity prediction, highlighting the most influential factors according to SHAP values. SHAP, Shapley additive explanations; GBM, Gradient Boosting Machine

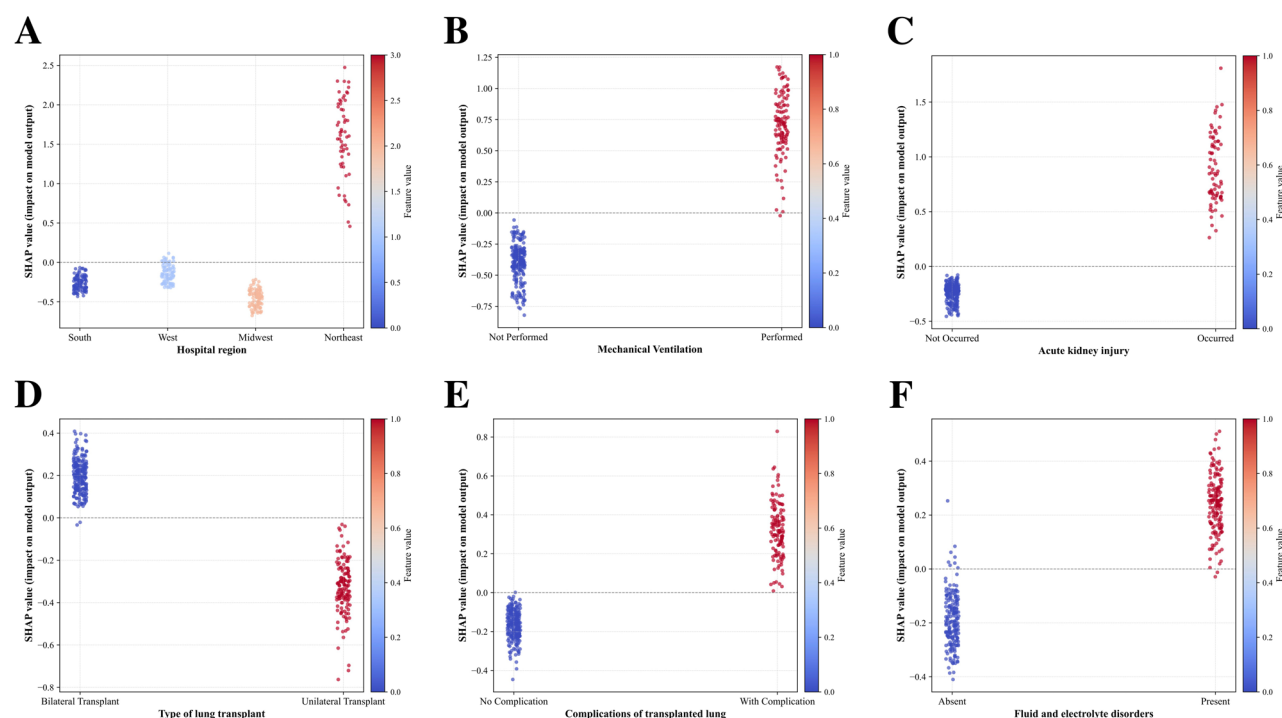


Fig. 5 SHAP-based local interpretation plots of the GBM model. Dependence plots for the six most influential features: **(A)** Hospital region, **(B)** Mechanical ventilation, **(C)** Acute kidney injury, **(D)** Type of lung transplant, **(E)** Complications of transplanted lung, and **(F)** Fluid and electrolyte disorders. Each point represents an individual in the test set, with colors indicating feature values. Positive SHAP values indicate higher possibility of predicting high hospital charges. SHAP, Shapley Additive Explanations; GBM, Gradient Boosting Machine

region, mechanical ventilation, acute kidney injury, lung transplant type, complications of transplanted lung, and fluid and electrolyte disorders. Hospital region is an important factor. The charges were significantly higher in northeastern region, which may be attributed to the high cost of healthcare practices and economic factors. Regional variation in healthcare spending is well

documented: New England and the Mideast (which include many Northeastern states) consistently exhibit among the highest per-capita health spending in the U.S., reflecting higher hospital prices and input costs in these regions [12]. Prior studies have shown that the cost of an ICU day for mechanically ventilated patients is significantly higher than for non-ventilated patients

[13, 14]. Additionally, prolonged ventilation increases the risk of complications, which further extends length of stay and drives up resource utilization [15]. Taken together, these findings support our model's identification of mechanical ventilation as a major contributor to inpatient charges among lung transplant recipients. Acute kidney injury (AKI) was another major predictor of high hospital charges. This aligns with substantial evidence showing that AKI—particularly when it occurs in postoperative or critically ill patients—dramatically increases morbidity, resource utilization, and total hospital costs. AKI is associated with longer ICU and hospital length of stay, greater need for renal replacement therapy, higher rates of complications, and increased mortality, all of which contribute significantly to overall expenditures [16]. Bilateral LT significantly increased the total hospital charges, which aligns with the procedural complexities and postoperative care requirements of bilateral LT. These patients often require more prolonged intensive care unit stays, more frequent interventions, and a higher level of postoperative care, all contributing to increased hospital costs. Complications are common after LT that worsen quality of life and contribute to high healthcare costs [17, 18]. Fluid and electrolyte disorders also emerged as a critical factor, indicating the additional medical management required to treat these conditions in the perioperative period. Managing fluid and electrolyte imbalances often requires continuous monitoring and additional treatments, which add to the overall hospital charges.

The predictive model developed in this study offers several practical implications. For clinicians, it provides a tool to identify patients at risk of incurring high hospital charges, enabling proactive management to mitigate patients' costs. Healthcare administrators can leverage the model to allocate resources more efficiently, ensuring that high-risk patients receive appropriate care without compromising financial sustainability. Policymakers could also use this model to inform decisions on funding allocations and to design interventions aimed at cost reduction in LT. While the NIS data used (2005–2014) are over a decade old, they remain valuable for characterizing charge drivers and resource use patterns within that historical period. At the same time, changes in payer mix, socioeconomic context, clinical practice, and hospital pricing over time may influence model performance, and thus application beyond 2014 should be interpreted with caution. Importantly, the large, nationally representative nature of the NIS supports rigorous machine learning and interpretability analyses, providing a foundation for subsequent application and methodological extension in related clinical and health services research settings.

Limitations

There were several limitations in the study. First, the machine learning model was developed and validated using NIS data from 2005 to 2014. Several predictors and the outcome definition are potentially time-varying, including primary payer, socioeconomic measures, and hospital charges, which may influence model calibration and the stability of estimates across different time periods. Second, the study included data from the NIS, which may not capture all relevant clinical variables. The NIS database includes only inpatient data, excluding outpatient costs and long-term follow-up, which are critical for a comprehensive cost analysis for LT patients. Third, hospital charges can vary widely based on institutional policies, insurance coverage differences, and individual patient preferences. Our model may not fully account for these variations, limiting its precision across different healthcare settings and patient populations. Fourth, lung transplantation cases were identified using ICD-9-CM procedure codes (33.51, 33.52) as recommended in prior HCUP-based studies. Although ICD-9-based phenotyping may introduce minor classification bias [19], these codes are the standard method for identifying lung transplantation hospitalizations in administrative databases. Because our primary goal was to analyze cost determinants within the lung transplant population rather than to estimate national incidence, any residual misclassification is unlikely to materially affect the study conclusions.

Conclusions

Our study demonstrates the potential of machine learning models in predicting high hospital charges, offering a valuable tool for improving the efficiency of LT for COPD patients. By integrating SHAP-based interpretation, our analysis identified modifiable factors—such as prolonged mechanical ventilation, acute kidney injury, complications, and fluid or electrolyte disturbances—that disproportionately contribute to higher hospital charges. These insights can guide clinicians to target perioperative optimization (e.g., early renal protection protocols, fluid management, respiratory weaning strategies) and hospital administrators to implement risk-adjusted resource planning. Furthermore, recognizing regional and institutional variations in costs may assist policymakers and payers in developing equitable reimbursement models and performance benchmarks for transplant centers. In this way, predictive modeling not only identifies high-cost patients but also delineates actionable pathways for reducing avoidable expenditures and improving post-transplant outcomes.

Abbreviations

HCUP	Healthcare Cost and Utilization Project
NIS	Nationwide Inpatient Sample

ICD-9-CM	International Classification of Diseases, Ninth Revision, Clinical Modification
APR-DRG	All Patient Refined Diagnosis Related Groups
LR	Logistic Regression
RF	Random Forest
GBM	Gradient Boosting Machine
XGBoost	Extreme Gradient Boosting
CatBoost	Categorical Boosting
LightGBM	Light Gradient Boosting Machine
SHAP	Shapley Additive Explanations
RFE	Recursive Feature Elimination
AUC	Area Under the Curve
ROC	Receiver Operating Characteristic
AP	Average Precision
P-R	Precision-Recall
LoS	Length of Stay
COPD	Chronic Obstructive Pulmonary Disease
ECMO	Extracorporeal Membrane Oxygenation
BMC	Bilateral Lung Transplantation
MICE	Multiple Imputation by Chained Equations

Supplementary Information

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Supplementary Material 1

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Author contributions

Q. D. and T. S. contributed equally to this work as co-first authors. X. Z. and T. S. contributed to the study conception and design. Material preparation, data collection and analysis were performed by Q. D. Q. D., M. T. S., Z. Z., Y. L., M. Z., P. B., and X. Z. The first draft of the manuscript was written by X. Z. and Q. D. All authors read and approved the final manuscript.

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Data availability

Research data will be shared upon request to the corresponding author.

Declarations

Ethics approval and consent to participate

The study was exempt from approval of the Institutional Review Board at the Eighth Medical Center, Chinese PLA General Hospital. The study was conducted in accordance with the Declaration of Helsinki. Informed consent to participate from each participant was waived as assessed by the Chinese PLA General Hospital Institutional Review Board.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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